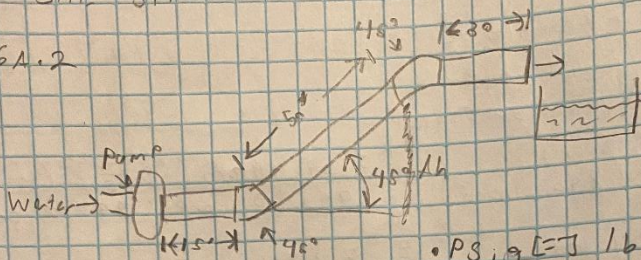


CHE 311

6A.2



$\rho = 62.4 \text{ lb/ft}^3$, $\mu = 1.002 \text{ cP}$, $\rho = 0.9982 \frac{\text{g}}{\text{mL}}$
 $D = 3.0688 \text{ in}$
 $w = 1.2 \text{ gal/min} \sim \text{mass flow rate}$
 $45^\circ \text{ elbows} \& 15 \text{ ft}$ additional pressure drops
 $w = \rho V \pi R^2 \Rightarrow V = w / \rho \pi R^2$
 $Re = \frac{\rho V D}{\mu} = \frac{w D}{\mu \pi R^2} = \frac{w D}{\mu (D/4) \pi}$
 $= \frac{4w}{\mu D \pi} = \frac{4(1.2 \frac{\text{gal}}{\text{min}})}{(1.002 \text{ cP})(3.0688 \text{ in}) \pi}$

$1 \text{ gal} = 3.7853 \text{ L} = 231 \text{ in}^3 = 0.13368 \text{ ft}^3$

$cP = 10^{-2} \frac{\text{g}}{\text{cm s}}$

$1 \text{ in} = 0.0254 \text{ m} = 2.54 \text{ cm}$

$\frac{4(1.2 \text{ gal/min})}{0.1002 \frac{\text{g}}{\text{cm s}} \cdot 7.74 \text{ cm} \cdot \pi} = \frac{1.2 \text{ gal}}{1.245 \text{ s}} = 4.89 \frac{\text{gal}}{\text{s}}$

Turbulent $Re = 18479 < 18.48 \frac{\text{L}}{\text{mL}} < 2000 \cdot 4.88 \frac{\text{gal}}{\text{mL}}$

$f = \frac{0.0001}{Re^{1/4}} = 0.0068$

$Fr = \frac{\Delta P}{\rho L} = (2 \pi R L) \left(\frac{1}{2} \rho V^2 \right) (0.0068)$
 $= \pi R^2 (P_0 - P_L)$ $L = 45 \text{ ft} (15 + 15) \text{ ft} = 289.5 \text{ cm}$
 $R = \frac{D}{2} = 230.72 \text{ cm}$

$-\Delta P = 2 \pi R L \cdot \frac{1}{2} \rho V^2 (0.0068) = \frac{11 \rho w^2 (0.0068)}{\pi^2 R^2 A (230.72 \text{ cm})}$

$\Delta P = \left(\frac{D}{2} \right)^4 = \frac{D^4}{16} = 230.72 \text{ cm}^4$

2A2)

• $1 \text{ lb} = 4.5359 \text{ g}$

• $f \text{ ft}^3 = 28.816 \text{ L} \rightarrow 22703 \text{ L}^3 = f \text{ ft}^3$

$1.022 \cdot 10^{-4} \frac{\text{ft}^3}{\text{s}^2} = -\Delta P$

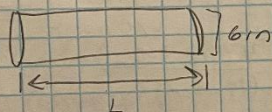
$\frac{L \cdot w^2 \cdot (0.068)}{\rho \pi^2 (899.21 \text{ cm}^3)} = \frac{3129.44 \text{ cm} \cdot (.8 \frac{\text{cm}}{\text{s}})^2 (0.068)}{998.2 \frac{\text{g}}{\text{cm}^3} \pi^2 (899.21 \text{ cm}^3)}$

$\therefore 3021 \rightarrow 1.130 \text{ L} = 3.10 \cdot 10^{-4} \frac{\text{L}^3}{\text{s} \cdot \text{cm}^4 \cdot \text{s}^2} = \Delta P$

$L = 1000 \text{ cm}^3$

$\frac{\text{cm}^9}{\text{s} \cdot \text{cm}^4 \cdot \text{s}^2} \rightarrow \frac{\text{cm}^5}{\text{s}^2}$

6A3)



• $L = 1320 \text{ ft}$ • $\Delta P = 0.25 \text{ psi} [=] \text{ lb/in}^2$

• $D = 6.00 \text{ in}$ • Hydraulically smooth $\rightarrow k/D = 0$

Obj: Find flow rate • $\rho = .998 \text{ g/mL}$

① $\Delta P = P_0 - P_L + \rho g h = .25 \frac{\text{lb}}{\text{in}^2} = 1723.75 \text{ Pa}$

$1 \text{ psi} [=] 6895 \text{ Pa}$

② $F_u = \frac{\rho R^2 [\Delta P]}{L f} = \frac{(\rho R^2) (\frac{4}{\pi} P \langle v \rangle^2) f}{L f} = \rho \langle v \rangle^2 = \rho \left(\frac{4w}{\pi D^2 \pi} \right)^2$

$w = \rho \langle v \rangle^2 R^2 \pi \rightarrow \langle v \rangle = w / (\rho (\frac{D}{2})^2 \pi) = \langle v \rangle$

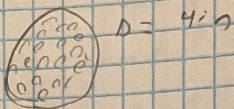
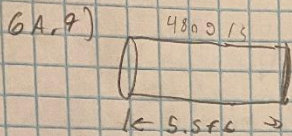
③ Assume turbulence

$\frac{1}{\sqrt{f}} = 1.56 \log \left[\frac{6.9 \mu}{D \rho \langle v \rangle} + 0 \right] \quad \mu = 1.002 \text{ cP}$
 $= 1.002 \cdot 10^{-2} \frac{\text{g}}{\text{cm} \cdot \text{s}}$

$\sqrt{\frac{R [\Delta P]}{L f \rho}} = \langle v \rangle = \sqrt{\frac{R [\Delta P]}{L \rho}} \cdot \left(1.56 \log \left[\frac{6.9 \mu}{D \rho \langle v \rangle} \right] \right)$

Would like to copy the example into Python (kg) but between dealing with units, writing the script, and limited time I'll leave it as is.

(3)



Glass sphere $D = 0.0625 \text{ in}$

$\epsilon = .41$

For con at 300°K : $\mu = 1.495 \cdot 10^{-4} \frac{\text{g}}{\text{cm}\cdot\text{s}}$

$P_L = 3 \text{ atm}$ $P_o = 25 \text{ atm} \rightarrow \Delta P = 22 \text{ atm}$

$\epsilon = \frac{\text{Volume of solid particles}}{\text{Volume of the empty column}}$

$$S_o = \pi R^2 L_o = \pi (16 \text{ in})^2 (66 \text{ in}) = 829.38 \text{ in}^3$$

$$.41 (829.38 \text{ in}^3) = S_a = 340.04 \text{ in}^3$$

$$\frac{4}{3} \pi \left(\frac{D}{2}\right)^3 = 1.278 \cdot 10^{-4} \text{ in}^3 \quad \text{volume for each sphere}$$

$$R_h = \frac{S_a}{Z} = \frac{\epsilon}{(1-\epsilon) \rho_p} = \frac{\epsilon D_p}{6(1-\epsilon)} = \frac{(.41)(0.0625 \text{ in})}{6(1-.41)}$$

$$R_h = .00694 \text{ in}$$

$$R_{Ch} = \frac{G_o 4 R_h}{\epsilon \mu} = \frac{(W) \mu (4 R_h)}{(S_o) \epsilon \mu} \quad \begin{matrix} \text{Guess} \rightarrow \text{Laminar} \\ \text{Tube} \end{matrix}$$

$$\frac{P_o - P_L}{L} = \frac{f_{\text{tube}} R_{Ch} \rho^2}{2 R_h} = \frac{f_{\text{tube}} G_o^2}{\rho \epsilon^2 (2 R_h)} = \frac{f_{\text{tube}} W^2}{S_o^2 \rho \epsilon^2 (2 R_h)}$$

Turbulent (Guss and) $f_{\text{tube}} = 7/12$

$$\frac{\Delta P}{L} = \frac{(.58) W^2}{(829.38)^2 \rho (.41)^2 (.01388)}$$

$\rho = 1.78 \text{ kg/m}^3$ at 300°K

$$.33 \frac{\text{atm}}{\text{in}} = \frac{(.58) W^2}{2856.82 \frac{\text{in}^6 \text{kg}}{\text{m}^3}}$$

$W [\text{kg}]$ mass/time
 $\frac{N}{m^2} [\text{kg} \cdot \frac{1}{\text{s}^2 \text{m}}]$

$$1625.44 \frac{\text{atm} \cdot \text{kg}}{\text{m}^3} = W^2 [\text{kg}^2]$$

$\frac{\text{atm} \cdot \text{kg}}{\text{m}^3} \rightarrow \frac{\text{kg}^2}{\text{s}^2 \text{m}^3}$

$$= 116954769 \frac{\text{kg}^2}{\text{s}^2} = W$$

$\frac{\text{kg}^2}{\text{s}^2} \rightarrow \frac{\text{kg}^2 \cdot \text{in}^5}{\text{s}^2 \text{m}^5}$

6B.2)

$$F_z = 4.4 \cdot 10^{-30} ; F_x = 2 \int_0^W \int_0^L \left(\mu \frac{\partial v_x}{\partial y} \right) dx dz$$

$$= 2 \int_0^W \int_0^L \left(\mu v_\infty \frac{df'}{d\eta} \cdot \frac{\partial \eta}{\partial y} \right)_{y=0} dx dz$$

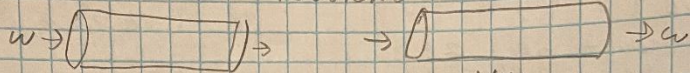
$$= 2 \int_0^W \int_0^L \left(\mu v_\infty f''(\eta) \cdot \sqrt{\frac{1}{2}} \frac{v_\infty}{\nu x} \right) dx dz$$

$$F_x = 1.328 \sqrt{\rho \mu L W^2 v_\infty^3}$$

close enough $\rightarrow 28 \Delta p_1 =$

6 B.S

Turbulent



Old
 $\bullet D = 2R$
 $\bullet L$

Smooth

$\bullet P_0 = P_L$
 $\bullet w$

New

$\bullet D/2 = 2R$
 $\bullet L$

Smooth

$\bullet P_0 = P_L$
 $\bullet w$

higher pressure needed

a) Eq. 6.2-12: $f_c = 0.0791 (2.1 \cdot 10^3 < Re < 10^5)$
 $Re = \frac{w D}{\mu}$

$\rho \langle v \rangle^2 \pi R^2 = w = \rho \langle v \rangle^2 \pi R^2 \rightarrow \langle v \rangle = \frac{w}{\pi R^2}$
 $\rho \langle v \rangle^2 \pi \left(\frac{D}{4}\right)^2 = \rho \langle v \rangle^2 \pi \left(\frac{D}{16}\right)^2 = \rho \langle v \rangle^2 \pi \left(\frac{D}{16}\right)^2 \Delta P$

$\frac{D^4}{16} = \langle v \rangle^2 \left(\frac{D^2}{4}\right) = \langle v \rangle^2 \left(\frac{D^2}{16}\right) \rightarrow \langle v \rangle^2 \cdot 4 = \langle v \rangle^2$

$Re = \frac{\rho \langle v \rangle D}{\mu} = \frac{4w}{\pi \mu D} \quad Re = \frac{\rho \langle v \rangle (D/4)}{\mu}$

$\frac{4w}{\pi \mu D} = \frac{w D}{\pi \left(\frac{D}{16}\right)^2 \mu} = \frac{64w}{\pi \mu D}$

$f_1 = \frac{0.0791}{\left(\frac{4w}{\pi \mu D}\right)^{1/4}} \quad f_2 = \frac{0.0791}{\left(\frac{8w}{\pi \mu D}\right)^{1/4}}$

$F_{K1} = \left(\frac{32 \mu L}{D^4}\right) \left(\frac{4w}{\pi}\right)^2 f_1 = \frac{32 \mu L}{D^4} \left(\frac{4w}{\pi}\right)^2 (P_0 - P_L)$

$f_1 = \frac{1}{2} \frac{\langle v \rangle^2 \rho}{\Delta P} \quad f_2 = \frac{1}{2} \frac{\langle v \rangle^2 \rho}{\Delta P}$

$\langle f_1 \rangle = \frac{w^2 L \cdot 2}{D \cdot \left(\frac{D}{16}\right)^2 \rho \Delta P} \quad \langle f_2 \rangle = \frac{w^2 L \cdot 2}{D \cdot \left(\frac{D}{16}\right)^2 \rho \Delta P}$

$w^2 = \frac{\rho \pi R^2 D^5 \Delta P}{32 \mu L} = \frac{\rho \pi R^2 D^5 \Delta P}{32 \mu L}$

$w^2 = \frac{\Delta P \left(\frac{4w}{\pi}\right)^2}{32 \cdot 0.0791 \mu} = \frac{\Delta P \left(\frac{8w}{\pi}\right)^2}{32 \cdot 0.0791 \mu} \quad 1.9 \Delta P = \frac{\Delta P \cdot 1.6}{1.624}$

Answer: 27X ← close enough → $28 \Delta P_1 = \Delta P_2$

6B.3

Result of 2B.3

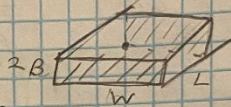
$w = 2BW \rho \langle v_z \rangle = - \frac{2W \rho \Delta P B^3}{3 \mu L}$

$$\langle \bar{v} \rangle = - \frac{\Delta P B^2}{3 \mu L}$$

Show that $f = 12/Re$ if

$$Re = (2B) \langle v_z \rangle \rho / \mu$$

for a thin slit of thickness $2B$ in laminar flow



$$F_R = A K f$$

$$= (2B L w) \left(\frac{1}{2} \rho \langle v_z \rangle^2 \right) f$$

$$F_R = (P_0 - P_L) (2B w)$$

$$f = \left(- \frac{\Delta P}{L} \right) \frac{2}{\rho \langle v_z \rangle^2} \rightarrow \frac{3 \mu w}{B^2} \cdot \frac{2B}{\rho \langle v_z \rangle^2} = \frac{6 \mu}{B^2 \rho \langle v_z \rangle}$$

$$f = \frac{12}{Re}$$

$$R_h = \frac{S}{Z} = \frac{2BW}{2(2B+w)} = \frac{2BW}{4B+2W} = \frac{BW}{2B+W}$$

$$Re_h = \frac{4 R_h \langle v_z \rangle \rho}{\mu} = \frac{4 \left(\frac{BW}{2B+W} \right) \langle v_z \rangle \rho}{\mu} = \frac{4BW \langle v_z \rangle \rho}{\mu (2B+W)}$$

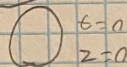
$$Re_h = \frac{4BW \langle v_z \rangle \rho}{(2B+W) \mu}$$

$$f = \frac{12}{Re_h} = \frac{12 (2B+W) \mu}{4BW \langle v_z \rangle \rho} = \frac{3 (2B+W) \mu}{BW \langle v_z \rangle \rho}$$

Answers

b) 1/c

68.7



$$\downarrow F_D = V_z^2$$

$$F_R = (F_n - F_s) + F_i = F_{form} + F_{friction}$$

$$F_R = (\pi R^2) \left(\frac{1}{2} \rho v_\infty^2 \right) f = \frac{4}{3} \pi R^3 \rho_s \rho_\infty - \frac{4}{3} \pi R^3 \rho g$$

$$f = \frac{409}{3 v_\infty^2} \left(\frac{\rho_{sea}}{\rho} - 1 \right) = .44 \quad \frac{\Delta z}{\Delta t} = V_z$$

$$\frac{409}{3 (.44)} \left(\frac{\rho_{sea}}{\rho} - 1 \right) = v_\infty^2 = v_z^2$$

$$F_R = (\pi R^2) \left(\frac{1}{2} \rho_{sea} .44 \right) 409 \left(\frac{\rho_{sea}}{\rho} - 1 \right) = \frac{4}{3} \pi R^3$$

$$= \pi R^3 g (\rho_{sea} - \rho)$$

$$\begin{aligned} r \frac{3 R g (\rho_{sea} - \rho)}{2 \Delta t \rho \left(\frac{\rho_{sea}}{\rho} - 1 \right)} &= \frac{3 R g (\rho_{sea} - \rho)}{2 \Delta t (\rho_{sea} - \rho)} \\ &= \frac{3}{2 \Delta t} = \frac{3}{4} \end{aligned}$$